Idemotency

1- Terraform plan → 0 changes

```
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Tas
k1/Terraform$ terraform plan
data.aws_ami.ubuntu: Reading...
aws_key_pair.deployer: Refreshing state... [id=bark-lab1-key]
aws_vpc.main: Refreshing state... [id=vpc-0501772616b46cd65]
data.aws_ami.ubuntu: Read complete after 1s [id=ami-04de259f26cbf4c1e]
aws_security_group.app: Refreshing state... [id=sg-0e8b10fe9cb4d0b82]
aws_internet_gateway.igw: Refreshing state... [id=igw-0046385c81e9ad3d7]
aws_subnet.public: Refreshing state... [id=subnet-0b887086a1722c047]
aws_route_table.public: Refreshing state... [id=rtb-0ff9f4d8d32eead26]
aws_security_group.db: Refreshing state... [id=sg-06ad05568e91d80a4]
aws_route_table_association.public: Refreshing state... [id=rtbassoc-0071be149420b211a]
aws_instance.app: Refreshing state... [id=i-068f406d70731457f]
aws_instance.db: Refreshing state... [id=i-05ff8de8c9ddc608c]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences,
so no changes are needed.
```

2- Ansible Playbook: 0 changes

```
TASK [app : Build app image] ***************************************************************************************************************************************************
ok: [app1]

TASK [app : Copy template app environment file to host] ************************************************************************************************************************
ok: [app1]

TASK [app : Copy template nginx config to host] ********************************************************************************************************************************
ok: [app1]

TASK [app : Template compose file] *********************************************************************************************************************************************
ok: [app1]

TASK [app : Wait for PostgreSQL on the database host] **************************************************************************************************************************
ok: [app1]

TASK [app : Start app] *********************************************************************************************************************************************************
ok: [app1]

PLAY RECAP *********************************************************************************************************************************************************************
app1                       : ok=17   changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
db1                        : ok=17   changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$
```

Non-root app, DB user, health checks, restart policies

```
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ KEY=~/.ssh/barq-lab
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ APP=$(terraform -chdir=Terraform output -raw app_public
_ip)
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ ssh -i $KEY ubuntu@$APP "sudo docker exec app id"
uid=10001(app) gid=10001(app) groups=10001(app)
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ ssh -i $KEY ubuntu@$APP "sudo docker ps --format '{{.Na
mes}} {{.Status}}'"
nginx Up 12 minutes (healthy)
app Up 13 minutes (healthy)
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ ssh -i $KEY ubuntu@$DB  "sudo docker exec postgres psql
 -U postgres -c '\du'"
                             List of roles
 Role name |                         Attributes
-----------+------------------------------------------------------------
 notesuser |
 postgres  | Superuser, Create role, Create DB, Replication, Bypass RLS

adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ ssh -i $KEY ubuntu@$DB  "sudo docker inspect -f '{{.Hos
tConfig.RestartPolicy.Name}}' postgres; systemctl is-enabled docker"
unless-stopped
enabled
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$
```

Create test data, Data survives DB container recreation

```
{"message":"user created"}
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducabili
ty/Barq-Task1$ curl -s http://$APP/users
[{"email":"A.hossammahmoud2164@nu.edu.eg","id":1,"username":"sdsds"},{"email":"gmail@gmail.c
om","id":5,"username":"sdsdsdsdsd"},{"email":"test1@x.com","id":6,"username":"test1"}]
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducabili
ty/Barq-Task1$ ssh -i $KEY ubuntu@$DB "cd /opt/postgres && sudo docker compose down && sudo
docker compose up -d --wait"
 Container postgres Stopping
 Container postgres Stopped
 Container postgres Removing
 Container postgres Removed
 Network postgres_default Removing
 Network postgres_default Removed
 Network postgres_default Creating
 Network postgres_default Creating
 Network postgres_default Created
 Network postgres_default Created
 Container postgres Creating
 Container postgres Created
 Container postgres Starting
 Container postgres Started
 Container postgres Waiting
 Container postgres Healthy
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducabili
ty/Barq-Task1$ curl -s http://$APP/users
[{"email":"A.hossammahmoud2164@nu.edu.eg","id":1,"username":"sdsds"},{"email":"gmail@gmail.c
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ curl -s http://$APP/users
[{"email":"A.hossammahmoud2164@nu.edu.eg","id":1,"username":"sdsds"},{"email":"gmail@gmail.com","id":5,"username":"sdsdsdsdsd"},{"email":"t
est1@x.com","id":6,"username":"test1"}]
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$
```

Both servers survive a reboot and containers restart automatically

```
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ cd ansible
&& ansible all -m ansible.builtin.reboot && cd ..
app1 | CHANGED => {
    "changed": true,
    "elapsed": 26,
    "rebooted": true
}
db1 | CHANGED => {
    "changed": true,
    "elapsed": 26,
    "rebooted": true
}
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ curl -s htt
p://$APP/users
[{"email":"A.hossammahmoud2164@nu.edu.eg","id":1,"username":"sdsds"},{"email":"gmail@gmail.com","id":5,"username":"sds
dsdsdsd"},{"email":"test1@x.com","id":6,"username":"test1"}]
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$
```

Cronjob, Backup → Delete → Restore Pattern works as expected

```
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ ssh -i $KEY ubuntu@$DB "sudo crontab -l"
#Ansible: daily postgres backup
0 2 * * * /usr/local/bin/pg_backup.sh >> /var/log/pg_backup.log 2>&1
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ ssh -i $KEY ubuntu@$DB "sudo /usr/local/bin/pg_backup.sh &&
sudo ls -lh /var/backups/postgres"
total 8.0K
-rw-r--r-- 1 root root 884 Oct 22 20:39 notesdb_2026-10-22_20-39-30.sql.gz
-rw-r--r-- 1 root root 884 Oct 22 20:43 notesdb_2026-10-22_20-43-28.sql.gz
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ ssh -i $KEY ubuntu@$DB "sudo docker exec postgres psql -U no
tesuser -d notesdb -c 'DELETE FROM users;'"
DELETE 3
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ curl -s http://$APP/users
[]
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ ssh -i $KEY ubuntu@$DB "sudo sh -c '/usr/local/bin/pg_restor
e.sh \$(ls -t /var/backups/postgres/*.sql.gz | head -1)'"
SET
SET
ALTER TABLE
adham@DESKTOP-1RF0NOI:/mnt/d/HomeLab-Personal-Projects/Barq Task 1 testing for reproducability/Barq-Task1$ curl -s http://$APP/users
[{"email":"A.hossammahmoud2164@nu.edu.eg","id":1,"username":"sdsds"},{"email":"gmail@gmail.com","id":5,"username":"sdsdsdsdsd"},{"email":"test1@x.com","id":6,"username
":"test1"}]
```